

MUGS-SLAM: Multi-Camera Gaussian Splatting SLAM

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1 Introduction

This project introduces MUGS-SLAM: an online multi-camera SLAM system that offers accurate pose tracking and generates photorealistic 3D reconstructions by exploiting 3D Gaussian Splats (**3DGS**). It is able to infer depth and normal maps for multiple cameras with a consistent per-frame scale. We demonstrated the capabilities of this system in real and synthetic datasets, using pinhole and fisheye camera models.

2 System Architecture

The proposed architecture starts with preprocessing the input frames of all cameras. These frames are sent to a DroidNet model for feature extraction at reduced resolution. A keyframe selector streams the high-confidence frames to a prior extraction module to obtain the corresponding depth and normal maps at full resolution. These keyframes undergo recurrent field transformations (**RFT**) and bundle adjustment (**BA**), together with histogram-based depth alignment, to ensure per-frame scale consistency and optimize the estimated pose.

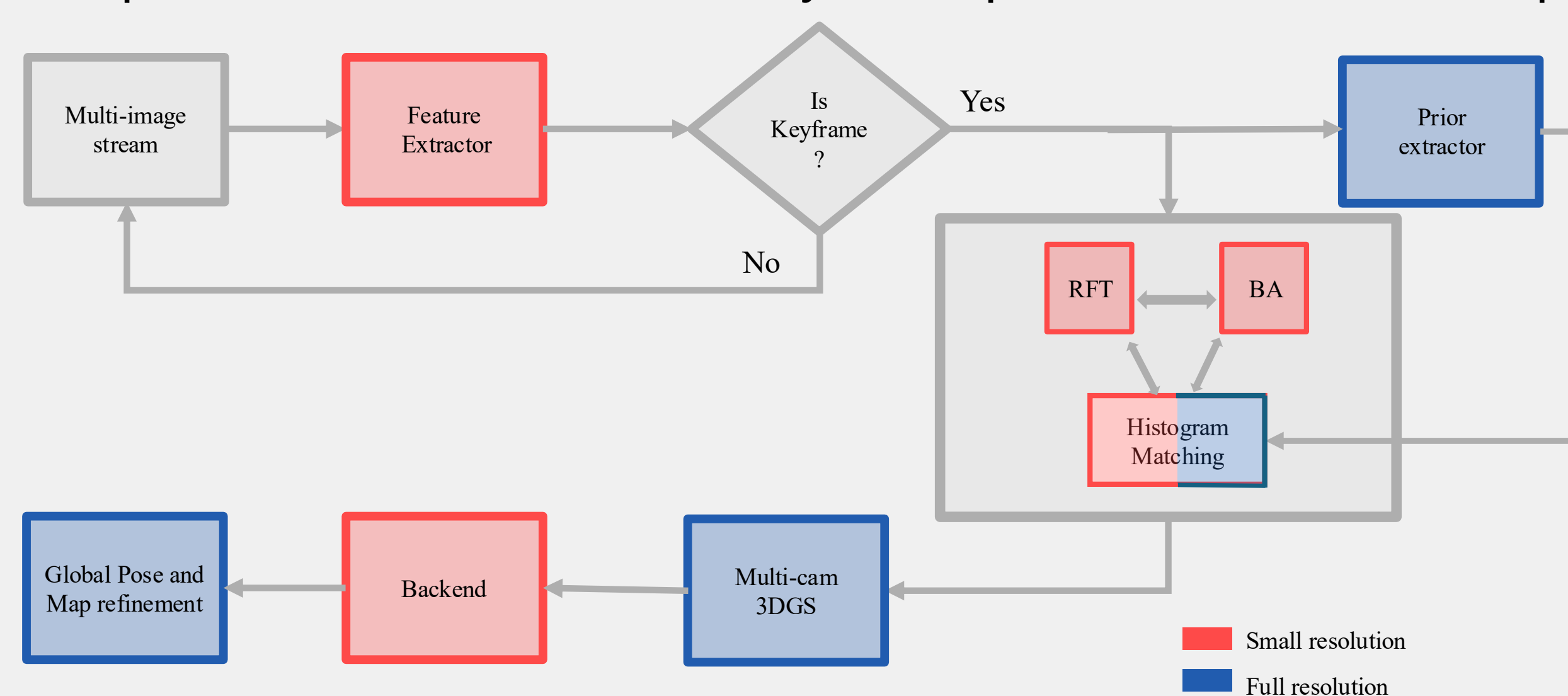


Figure 1: Flow-chart of the MUGS-SLAM system.

The aligned priors are sent to the 3DGS module to generate photo-realistic 3D reconstructions. The estimated camera poses are then iteratively adjusted to ensure track consistency and reconstruction fidelity. Finally, new keyframes are added to improve on-spot rotations, and global pose and map refinement is performed.

3 Materials and Methods

The MUGS-SLAM framework employs 3DGS as the core scene representation, enabling fast rendering and map optimization. For robust pose estimation, the BA module jointly optimizes camera poses, while the RFT provides reliable feature correspondences in challenging environments. To improve the alignment between predicted depths and Gaussian splats, MUGS-SLAM uses the Joint Depth and Scale Alignment (**JDSA**) algorithm together with histogram matching.

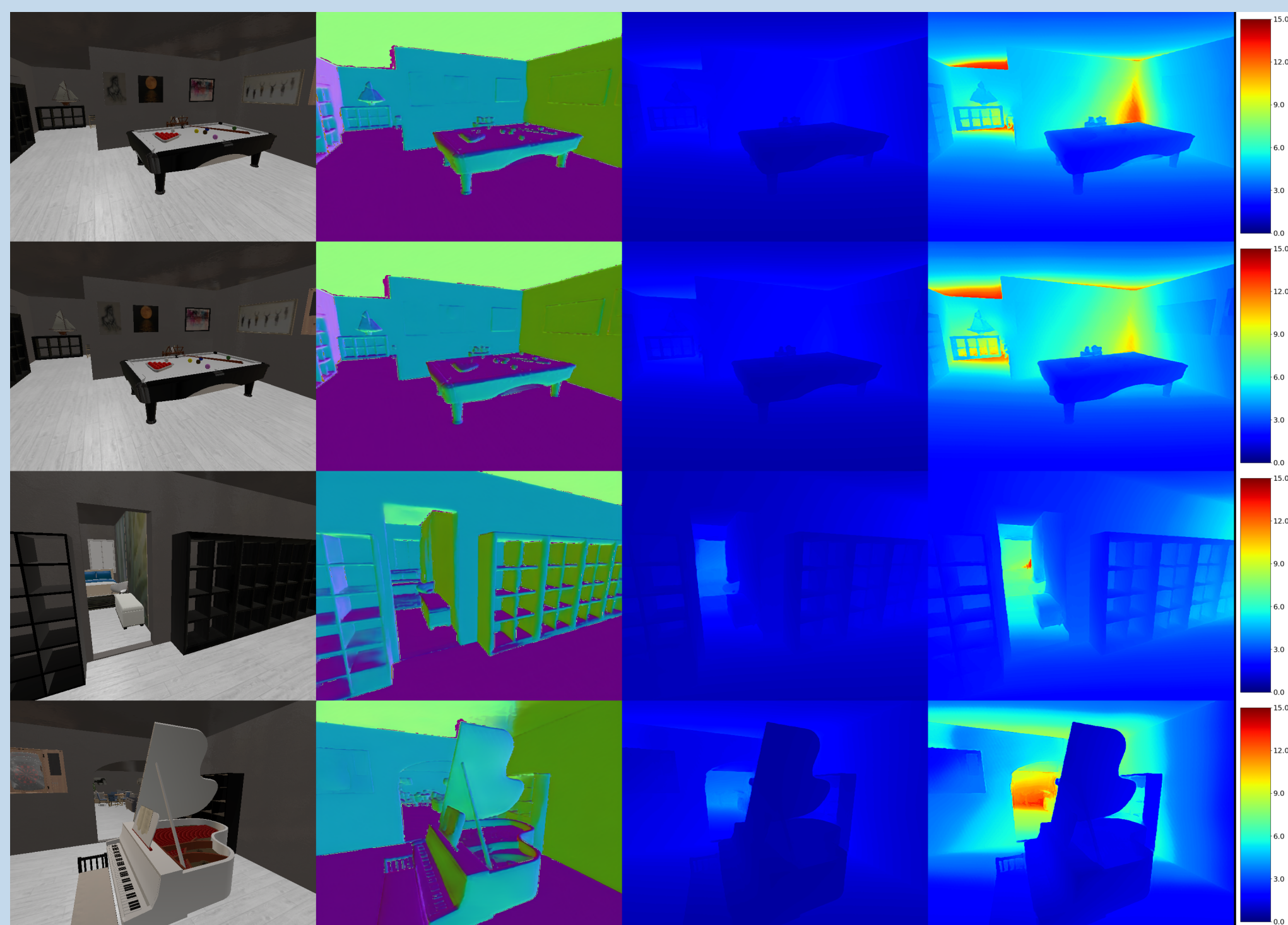


Figure 2: RGB inputs, prior normals and depths, and histogram-aligned depths.

4 Results and Discussion

The tracking and reconstruction performance of MUGS-SLAM on three datasets (Habitat, M2DGR, and Hilti) is summarized in Table 1.

Dataset	PSNR	SSID	APE [m]
Habitat	33.8	0.95	0.085
M2DGR	22.5	0.77	0.137
Hilti	23.9	0.86	0.096

Table 1: Performance of MUGS-SLAM on three synthetic datasets.

Figure 3 showcases the result of the 3DGS reconstruction using multi-camera inputs, generated from the dataset Room 01.



Figure 3: 3DGS reconstruction of Room 01 using the MUGS-SLAM system.

Lastly, Figure 4 presents the aligned pose error of the estimated trajectory using the multi-camera setup of the dataset Hilti Exp04.

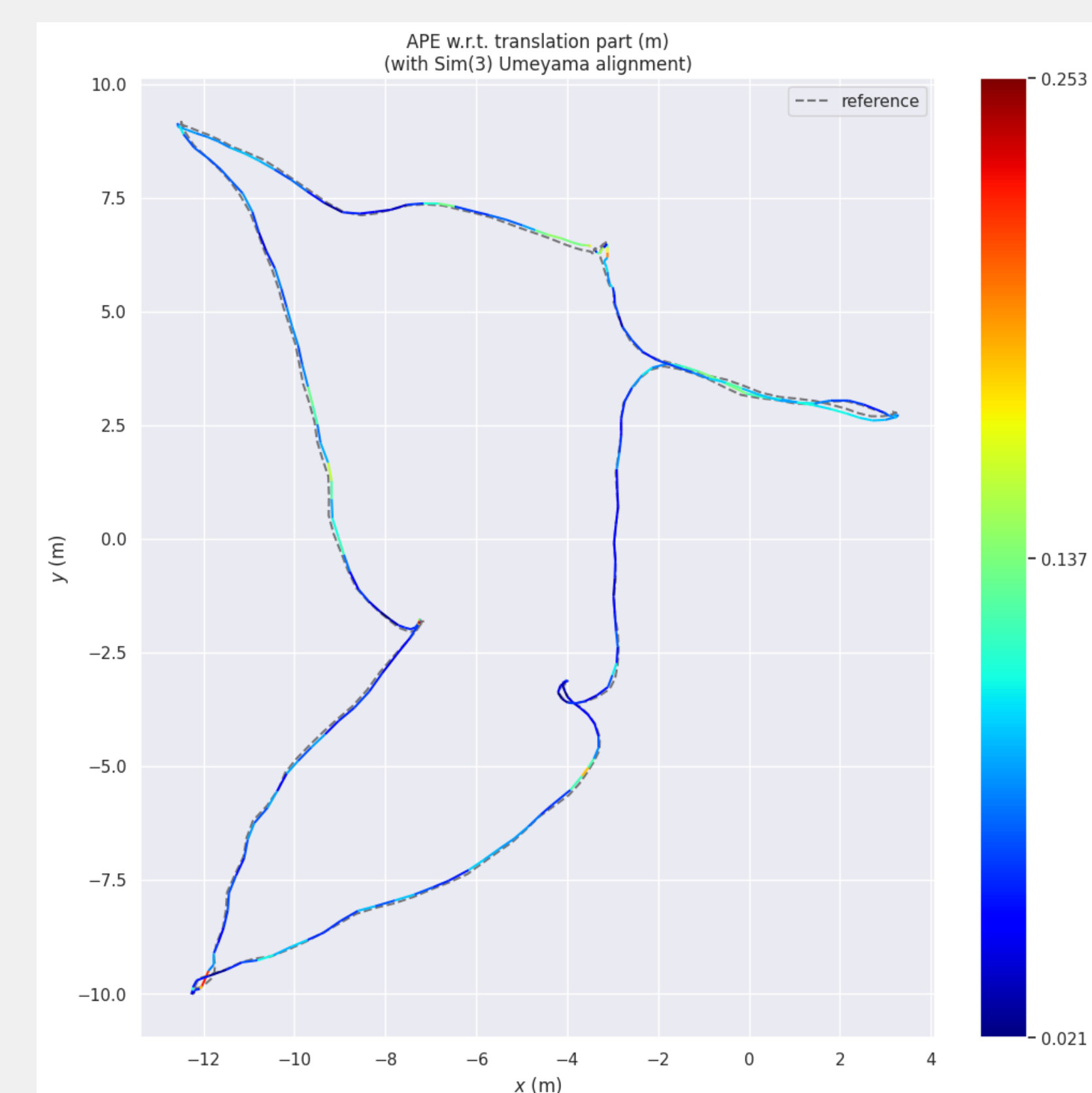


Figure 4: Aligned pose error of the estimated trajectory in Hilti Exp04.

5 Contributions

The main contributions of this work are:

- 1. Multi-camera Gaussian Splatting:** Adapted the single camera framework to build a *single* map for multi-camera setups.
- 2. Histogram Matching:** Significantly improved JDSA by introducing a novel way of aligning the depth prior to the full-resolution image.
- 3. CUDA kernels for Fisheye and Pinhole:** Implemented new kernels to parallelize forward and inverse projections, undistortion, and bundle adjustment for Fisheye and Pinhole models.
- 4. Custom dataset with Habitat-Sim:** Adapted the simulator to record multi-camera sequences for training and evaluation.

References

- [1] Wei Zhang, Qing Cheng, David Skuddis, Niclas Zeller, Daniel Cremers, and Norbert Haala. Hi-slam2: Geometry-aware gaussian slam for fast monocular scene reconstruction, 2024.
- [2] Wei Zhang, Sen Wang, Xingliang Dong, Rongwei Guo, and Norbert Haala. Bamf-slam: Bundle adjusted multi-fisheye visual-inertial slam using recurrent field transforms, 2023.